FORWARD DEPLOYED AI ENGINEERING

Capstone Project

Stage Five: Revision Log

Recording what your requirements got wrong and why

You are required to revise your requirements document at least once during the build. This is where you record what changed and what prompted the change.

WEEK TWO INTO WEEK THREE · COMPULSORY

Why this is compulsory

A requirements document written before any code exists is a set of educated guesses. Some of those guesses will be wrong, and building is how you find out which ones. A student that reaches the end of three weeks with an unchanged PRD has either been extraordinarily lucky or has stopped consulting the document, and the second is far more common than the first.

This log is not a confession. It is evidence that you were paying attention. The students who score well here are the ones who can point to a specific thing they discovered while building, explain how it contradicted an assumption they had written down, and show what they changed as a result.

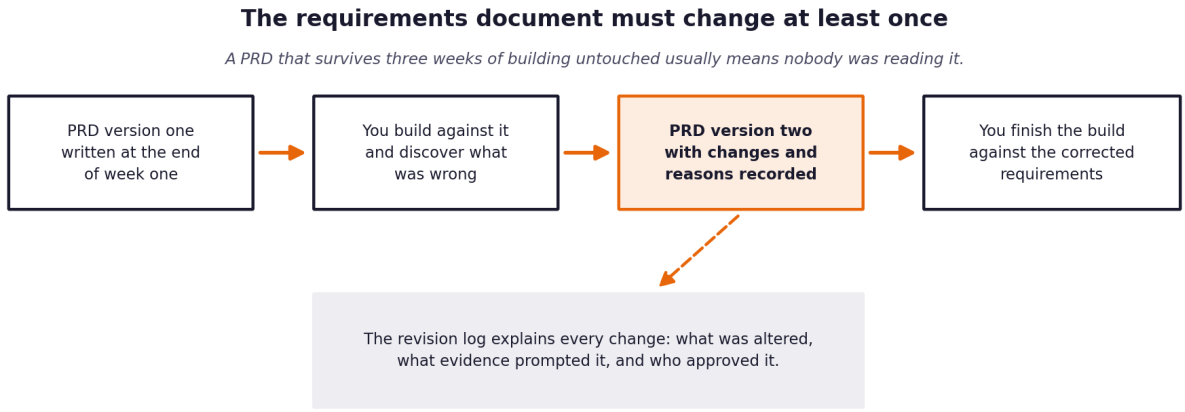


Figure 1. The revision sits in the middle of the build, not at the end.

What a weak revision log looks like

Vague entries such as tidied up wording or clarified requirements tell an assessor nothing. A strong entry names the requirement, states what it said before, states what it says now, identifies the specific thing that prompted the change, and says who agreed to it.

1 Section one: revision summary

Field	Value
New version number	2.0
Date of revision	2026-11-02 (submission preparation; confirm actual revision date)
Owner	Kaushik Dutta
Status	Draft pending student confirmation
Number of requirements changed	
Number of requirements added	
Number of requirements removed	

2 Section two: the changes

One row per change. Be specific about the trigger; that column is what distinguishes a considered revision from a tidying exercise.

Requirement ID	What it said in version one	What it says now	What prompted the change	Agreed by
FR-03	Predict urgency for routing.	Return separate urgency score; never treat low confidence as reliable priority; keep human triage.	Visible validation urgency accuracy 43.75%.	Student confirmation required
FR-05	Answer docs-supported tickets.	Auto-answer only if confidence, grounding and safety gates pass; otherwise escalate.	Only 8/80 auto-responded; 7/15 drafts blocked by lexical grounding.	Student confirmation required

NFR-03	High answer quality.	Require two independent human assessors on >=50 answers before safety claims.	Automated lexical guardrails cannot establish hallucination/citation accuracy.	Student confirmation required
NFR-04	Confidence threshold >=0.55.	Retain threshold as conservative gate; calibrate on separate held-out data before changing.	ECE is 16.6pp, above <=5pp target.	Student confirmation required
FR-09	Evaluate visible validation.	Harness must accept external input/output paths and support hidden 120 tickets without schema assumptions.	Submission gate runs on unseen data; hidden set unavailable locally.	Student confirmation required
NFR-05	Fairness <5pp.	Measure response quality by group using human ratings and sufficient n; treat historical FCR as context only.	Validation group n small; outcome gap not generated quality.	Student confirmation required
NFR-10	Use optional LLM for response.	Provider is optional and must fail closed; document live provider testing status.	Provider-free fallback needed; live provider not verified.	Student confirmation required

3 Section three: assumptions that turned out to be wrong

Go back to the assumptions section of your version one PRD. For each assumption, say whether it held. The ones that did not are the most interesting part of your report.

Assumption from version one	Did it hold?	What you found instead	What you changed because of it
Docs cover many tickets	Partly supported	71.4% dev answerability; validation expected doc hit 96.2% among answerable tickets with expected sources. Not proof of answer quality.	Keep retrieval; require grounded response and human review.
Urgency model is accurate enough	No	43.75% urgency accuracy on validation.	Do not automate priority; revise risk control.
Confidence score indicates calibrated certainty	No	ECE 16.6 percentage points.	Do not claim calibrated confidence; evaluate/calibrate separately.
Automated grounding means low hallucination	No	7/15 drafts blocked; lexical proxy is not human hallucination review.	Require independent human review before safety claims.
Historical records can show customer-time impact	No	No first reply timestamps; resolution duration is not reply latency.	Instrument live events before impact claims.
Non-fluent group has worse historic satisfaction	Not supported in development data	FCR 45.8% vs 43.2%; CSAT 3.04 vs 2.95. Validation is small and historical FCR differed.	Report mixed evidence; audit response quality by fluency.

4 Section four: what you decided not to change

Some things will look wrong during the build and turn out to be right, or right but not worth the cost of changing. Recording those decisions is as valuable as recording the changes.

What looked wrong	Why you left it	What it would have cost	Revisit when?

		to change	
60% FCR target vs brief's 65% variant	Use main Project Instructions/Evaluation Framework target; disclose disagreement.	Could create inconsistent evaluation if silently changed.	Ask course owner/assessor.
2-hour reply SLA vs <5 minute automation goal	They measure different things: contractual human SLA vs automated response target.	Conflating them would overstate impact.	Define live timestamp metrics.
Threshold 0.55 despite high ECE	Changing after visible validation risks tuning to set; keep as configured and disclose.	Could improve coverage superficially but compromise honest evaluation.	Use separate calibration split and safety review.
Provider integration not live-verified	No credentials/provider response evidence available.	A live demo could fail.	Student may configure a key and test separately.

5 Section five: the reflection

Answer these in full sentences. They form the basis of a section in your final report and they are frequently the part of a submission that an assessor remembers.

Question	Your answer
What did you most misunderstand about the problem when you wrote version one?	Student reflection required. Prompt: compare your actual initial beliefs with the evidence and explain one consequential revision.
Which piece of discovery work would have caught it earlier?	Student reflection required; use your own experience, not an inferred account.
What would you do differently next time?	Student reflection required.
What decision are you most confident defending, and why?	Student reflection required.